Software Requirements Specification

for

Measurement Process Simulator

Version 1.0 approved

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.

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1. Introduction

1.1 Purpose

This document defines the software requirements for a Java Swing-based wizard application designed to simulate the ISO/IEC 15939 software measurement process.

1.2 Scope

The application facilitates user profiling, quality scenario definition, automated metric scoring, and comprehensive performance analysis.

2. Overall Description

2.1 Product Perspective

A standalone desktop application built using the Model-View-Controller (MVC) architectural pattern to ensure clean separation between logic and UI.

2.2 Product Functions

User identification and validation.

- Scenario-based quality modeling (Health/Education).
- Linear normalization of raw quality data.
- Visual gap analysis and reporting.

2.3 Constraints

The system must be developed using only standard Java SE libraries. All measurement data and scenarios are hard-coded.

3. Functional Requirements

3.1 User Profiling

The system shall collect and validate the user's name, school, and session name before proceeding to the next step.

3.2 Scenario Definition

The system shall allow users to select between "Product/Process" types and "Health/Education" modes to load specific quality scenarios.

3.3 Automated Scoring

The system shall automatically calculate scores (1.0 - 5.0) for each metric based on their direction (Higher is Better / Lower is Better).

3.3 Performance Analysis

The system shall calculate weighted averages for each quality dimension and identify the lowest-performing area for **Gap Analysis**.

4. Non-functional Requirements

4.1 Usability (UI)

The interface shall utilize a **CardLayout** to implement a wizard-style workflow, featuring a progress indicator for user guidance.

4.2 Reliability

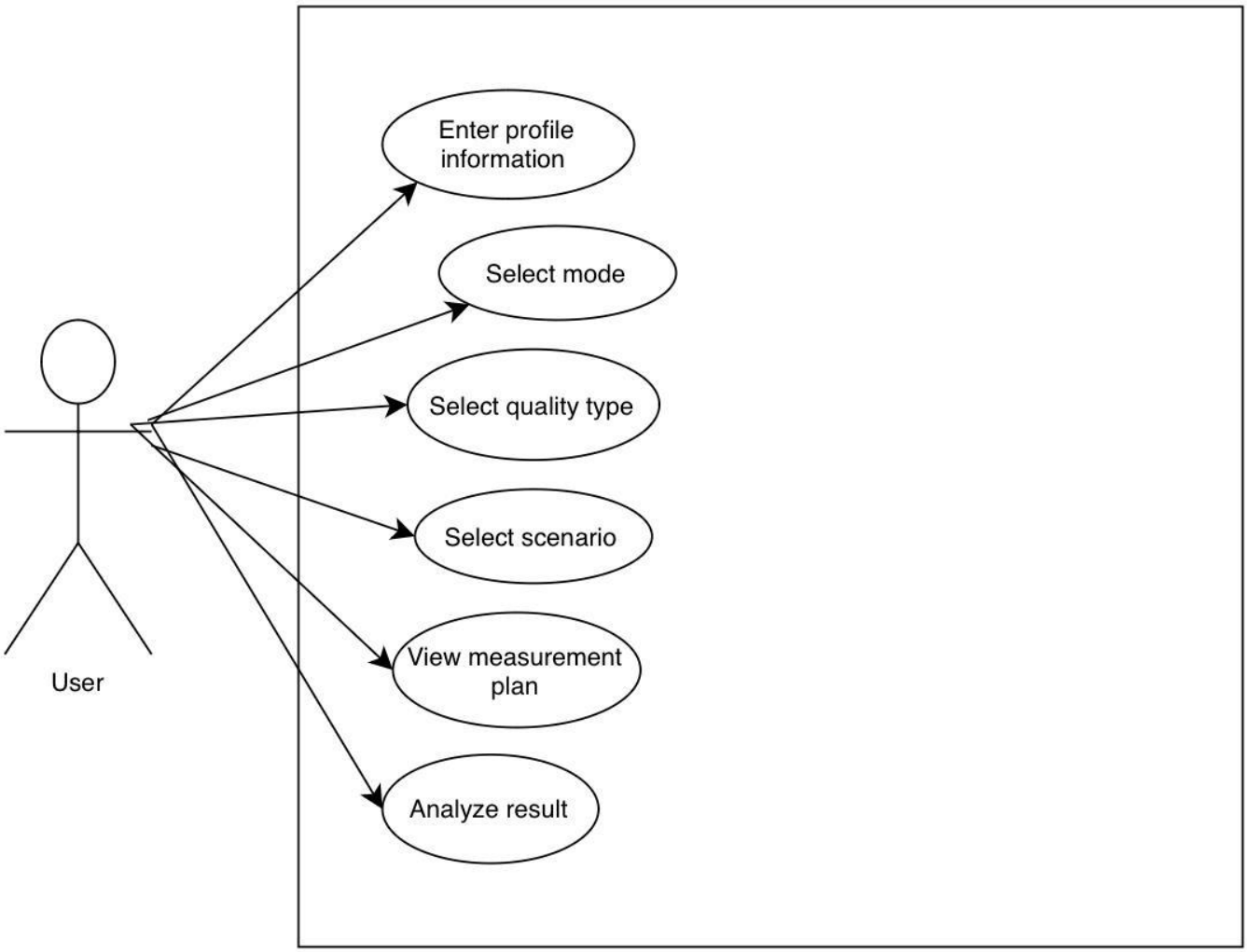
The system shall perform input validation at each step to prevent data loss or incomplete profiles.

4.3 Maintainability

The software shall be modularized into distinct Model, View, and Controller packages to facilitate future updates.

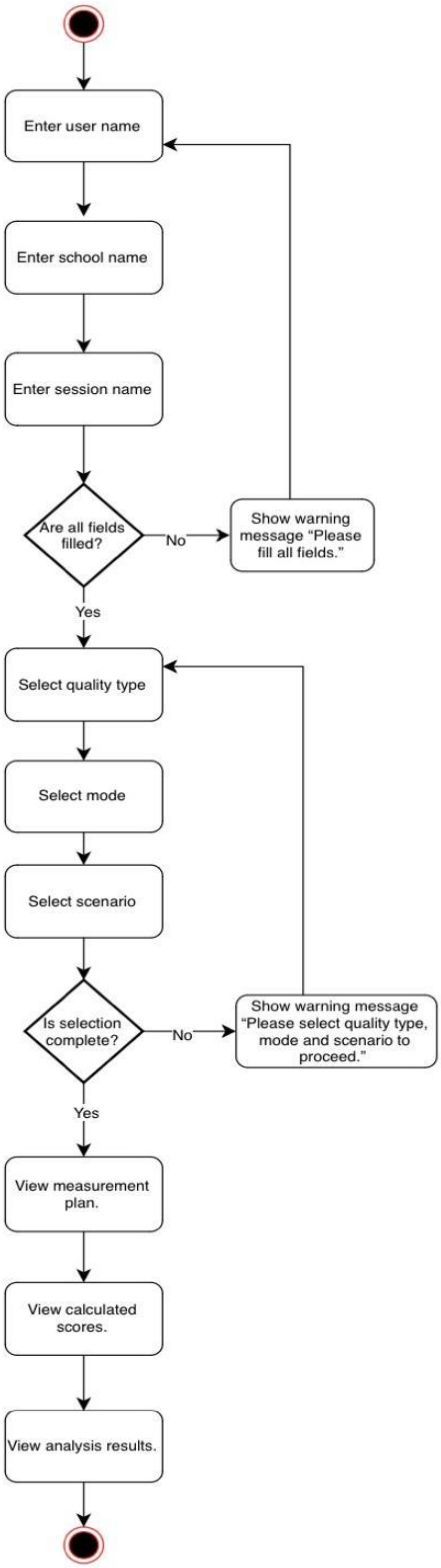
5. System Design (Analysis Models)

5.1 Use Case Diagram



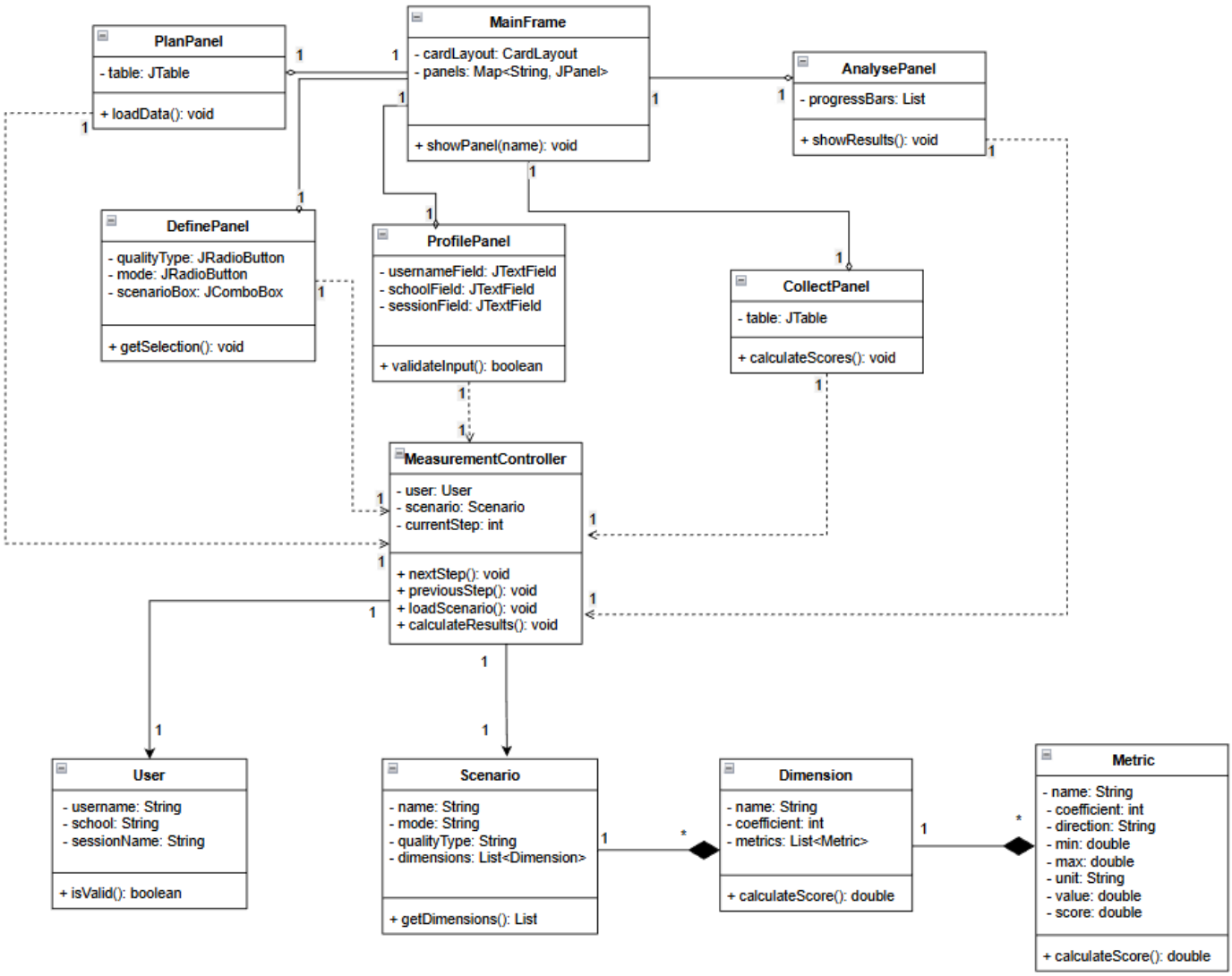
This diagram illustrates the interactions between the **User** and the **System**, specifically focusing on the goal-driven actions such as defining quality parameters and viewing analysis results.

5.2 Activity Diagram



This model represents the dynamic flow of the application, showing the sequence of steps from Step 1 (Profile) to Step 5 (Analysis) including decision points for data validation.

5.3 Class Diagram



This diagram defines the static structure of the system, showcasing the **MVC relationship**, class attributes, and the composition of Scenarios, Dimensions, and Metrics.